Unit 4: AP Free Response Practice #1 [2018 #1, shortened, 7 points]

$$Na_2S_2O_3(aq) + 4\ NaOCl(aq) + 2\ NaOH(aq) \rightarrow 2\ Na_2SO_4(aq) + 4\ NaCl(aq) + H_2O(l)$$

1. A student performs an experiment to determine the value of the ethalpy change, ΔH°_{rxn}, for the oxidation-reduction reaction represented by the balanced equation above.

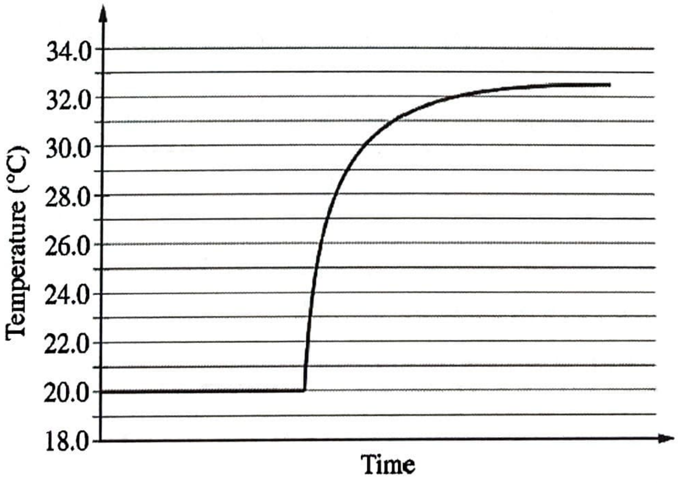
In the experiment, the student uses the solutions shown in the table below.

Solution	Concentration (M)	Volume (mL)
$Na_2S_2O_3(aq)$	0.500	5.00
$NaOCl(aq)$	0.500	5.00
$NaOH(aq)$	0.500	5.00

a. Using the balanced equation for the oxidation-reduction reaction and the information in the table above, determine which reactant is the limiting reactant. Justify your answer. [*1 point*]

(you will learn this next semester!)

The solutions, all originally at 20.0°C, are combined in an insulated calorimeter. The temperature of the reaction is monitored, as shown in the graph below.



b. According to the graph, what is the temperature change of the reaction mixture? [*1 point*]

$\Delta T = T_f - T_i = (32.5 - 20.0) = 12.5°C$

c. The mass of the reaction mixture inside the calorimeter is 15.21 g.

i. Calculate the magnitude of the heat energy, in joules, that is released during the reaction. Assume that the specific heat of the reaction mixure is 3.94 J/(g °C) and that the heat absorbed by the calorimeter is negligible. [*1 point*]

$q_{sys} = mC\Delta T = 15.21 \cdot 3.94 \cdot 12.5 = \underline{749\,J}$

ii. Using the balanced equation for the oxidation-reduction reaction and your answer to part (a), calculate the value of the enthalpy change of the reaction, $\Delta H°_{rxn}$, in kJ/mol_{rxn}. Include the appropriate algebraic sign with your answer. [*2 points*]

$n(NaOCl) =$ ~~[illegible]~~ $0.005\,L \times 0.500\,M = 0.00250\,mol\ NaOCl$

$\dfrac{0.00250\ NaOCl}{} \Big/ \dfrac{1\ RXN}{4\ NaOCl} = 0.000625\ mol\ rxn$

exo!

$\Delta H = \dfrac{q_{sys}}{n_{rxn}} = \dfrac{-0.749\,kJ}{0.000625\,mol} = -1200\ kJ/mol\ rxn$

The student repeats the experiment, but this time doubling the volume of each of the reactants, as shown in the table below.

Solution	Concentration (*M*)	Volume (mL)
$Na_2S_2O_3(aq)$	0.500	10.0
$NaOCl(aq)$	0.500	10.0
$NaOH(aq)$	0.500	10.0

d. The magnitude of the enthalpy change of the reaction, $\Delta H°_{rxn}$, in kJ/mol_{rxn}, calculated from the results of the second experiment is the same as the result calculated in part (c)(ii). Explain this result. [*1 point*]

ΔH_{rxn} is intensive & not dependent on moles present. If the volume doubled, q would double as well making the ratio q/n the same.

e. Write the balanced net-ionic equation for the given reaction. [*1 point*]

$S_2O_3^{-2} + 4OCl^- + 2OH^- \rightarrow 2SO_4^{-2} + 4Cl^- + H_2O$

(note: I just dropped out the spectators)
Na^+

Unit 4: AP Free Response Practice #2 [2017 #2, shortened, 5 points]

2. Answer the following questions about the isomers fulminic acid and isocyanic acid.

Fulminic acid can convert to isocyanic acid according to the equation below.

$$HCNO(g) \rightleftarrows HNCO(g)$$

fulminic acid *isocyanic acid*

Fulminic Acid	Isocyanic Acid
H—C≡N—Ö:	H—N̈=C=Ö:

a. Using the Lewis electron-dot diagrams of fulminic acid and isocyanic acid shown in the boxes above and the table of average bond enthalpies below, determine the value of ΔH° for the reaction of HCNO(g) to form HNCO(g). [*2 points*]

Bond	Enthalpy (kJ/mol)	Bond	Enthalpy (kJ/mol)	Bond	Enthalpy (kJ/mol)
N–O	201	C=N	615	H–C	413
C=O	745	C≡N	891	H–N	391

ΔH = bonds broken − bonds formed

$= (H\text{-}C + C\equiv N + N\text{-}O) - (H\text{-}N + N{=}C + C{=}O)$

$= (413 + 891 + 201) - (391 + 615 + 745)$

$= -246$ kJ/mol

b. A student claims that ΔS° for the reaction is close to zero. Explain why the student's claim is accurate. [*1 point*]

Accurate b/c there is no change of phase & no change in the numbers of moles between reactant & product. Therefore there will be a similar entropy, ΔS, for each.

c. Which species, fulminic acid (HCNO) or isocyanic acid (HNCO), is present in high concentration at equilibrium at 298 K? Justify your answer in terms of the thermodynamic favorability and the equilibrium constant. [*2 points*]

$\Delta G = \Delta H - T\Delta S = -246 - 298(0) \sim -246$ kJ/mol

because ΔG is negative, the forward reaction is favorable & more Isocyanic acid will be present.

Unit 4: AP Free Response Practice #3 [LTF Free Response #2, 10 points]

$$N_2(g) + 3\,H_2(g) \leftrightarrow 2\,NH_3(g) \qquad \Delta G^\circ_{rxn} = -34.1\ \text{kJ mol}^{-1}$$

3. The following questions relate the the synthesis reaction represented by the chemical equation above.

   a. Is the reaction thermodynamically favorable or unfavorable under standard conditions at 298 K? Justify your answer. [*1 point*]

Favorable! $\Delta G < 0$

   b. In terms of the equilibrium constant, K, for the above reaction at 25°C

      i. Predict whether K will be greater than, less than, or equal to one. Justify your choice. [*1 point*]

$K > Q$

      ii. Calculate its value. [*2 points*]

$\Delta G = -RT\ln K$

$-34.1 = -8.314 \cdot 298 \cdot \ln k$

$\dfrac{-34.1}{\left(-8.314 \cdot 298 / 1000\right)} = \ln k$

$K = 9.49 \times 10^5$

   c. Given the following data, determine the ΔH° for the above reaction. [*2 points*]

Substance	ΔH°_f (kJ mol^{-1})
$NH_3(g)$	-46.1

$\Delta H = \Sigma \Delta H_f(P) - \Sigma \Delta H_f(R)$

$= 2 \cdot -46.1 = \boxed{-92.2\ \frac{kJ}{mol}}$

   d. In terms of the standard entropy change, ΔS°

      i. Predict the sign of ΔS° for the above reaction. Justify your answer. [*1 point*]

$\Delta S < 0$ b/c 4 moles of reactants become 2 mol products.

ii. Calculate the value of ΔS° for the above reaction at 25°C. [*1 point*]

$\Delta G = \Delta H - T\Delta S$

$-34.1 = -\cancel{46.1} + T\Delta S$

298

-92.2

$\Delta S = 0.195\ kJ/mol \cdot K = 195\ J/mol \cdot K$

e. Using the data in the table below and the enthalpy of reaction, ΔH°_{rxn} determined in part (c), calculate the approximate bond energy of the nitrogen-hydrogen bond in ammonia. [*2 points*]

Bonds	Approximate Bond Energy ($kJ\ mol^{-1}$)
N – H	???
H – H	430
N ≡ N	960

$\Delta H_{rxn} = \Sigma$ bonds broken $- \Sigma$ bonds formed

$-92.2 = (N{\equiv}N + 3\ H{-}H) - (6\ N{-}H)$

$-92.2 = (960 + 3 \cdot 430) - 6(N{-}H)$

$(N{-}H) = 390\ kJ/mol$

Unit 4: AP Free Response Practice #4 [2013 FR #3, 10 points]

$$MgO(s) + 2\,H^{+}(aq) \rightarrow Mg^{2+}(aq) + H_2O(l)$$

4. A student was assigned to the task of determining the enthalpy change for the reaction between solid MgO and aqeous HCl represented by the net ionic equation above. Ther student uses a polystyrene cup calorimeter and performs four trials. Data for each trial are shown in the table below.

Trial	Volume of 1.0 *M* HCl (mL)	Mass of MgO(*s*) Added (g)	Initial Temperature of Solution (°C)	Final Temperature of Solution (°C)	ΔT
~~1~~	~~100.0~~	~~0.25~~	~~25.5~~	~~26.5~~	1.0
2	100.0	0.50	25.0	29.1	4.1
3	100.0	0.25	26.0	28.1	2.1
4	100.0	0.50	24.1	28.1	4.1

a. Which is the limiting reactant in all four trials, HCl or MgO? Justify your answer. [*1 point*]

MgO b/c as MgO increased, heat evolved increased.

b. The data in one of the trials is inconsistent with the data in the other three trials. Identify the trial with the inconsistent data and draw a line through the data from that trial in the table above. Explain how you identified the inconsistent data. [*1 point*]

Trial 1 → the ratio of ΔT to mass of MgO was not consistent with the other trials

For parts (c) and (d), use the data from one of the other three trials (i.e., not from the trial you identified in part (b) above.) Assume the calorimeter has a negligible heat capacity and that the specific heat of the contents of the calorimeter is 4.18 J/(g °C). Assume that the density of the HCl(aq) is 1.0 g/mL.

c. Calculate the magnitude of q, the thermal energy change, when the MgO was added to the 1.0 M HCl(aq). Include units with your answer. [*2 points*]

Trial 2: $q = mC\Delta T = (100 + 0.50) \cdot 4.18 \cdot 4.1$

$= 1722 \rightarrow 1700\text{ J}$

d. Determine the student's experimental value of ΔH° for the reaction between MgO and HCl in units of kJ/mol_{rxn}. [*2 points*]

$$\Delta H = q/n = -1700 / 0.0124$$

$$0.50g \,/\, \frac{1 mol}{40.31g} = 0.0124 mol$$

$$= -137\ kJ/mol$$

e. Enthalpies of formation for substances involved in the reaction are shown in table below. Using the informationin the table, determine the accepted value of ΔH° for the reaction between MgO(s) and HCl(aq). [*2 points*]

Substance	ΔH_f° (kJ/mol)
$MgO(s)$	−602
$H_2O(l)$	−286
$H^+(aq)$	0
$Mg^{2+}(aq)$	−467

$$\Delta H_{rxn} = \Sigma \Delta H_f (P) - \Sigma \Delta H_f (R)$$
$$= (-467 + -286) - (-602 + 2 \cdot 0)$$
$$= -151\ kJ/mol$$

f. The accepted value and the experimental value do not agree. If the calorimeter leaked heat energy to theenvironment, would it help account for the discrepancy between the values? Explain. [*1 point*]

Yes. if some heat was leaked to the environment, the ΔT would be smaller, making q an subsequently the experimental ΔH_{rxn} ~~smaller~~ less than the theoretical value.